



DVN36104 SPR 2026

Data Visualisation and Narratives
Spring 2026
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36104 · DATA VISUALISATION AND NARRATIVES

Vision, Color and Perception

Light, eyes, measurement, and screens

What survives as colour moves from the world into a visualisation?

One word, several different things

When we say “yellow” we might mean

spectral light	power concentrated near one wavelength
paint or flower	a surface reflectance under an illuminant
appearance	an observer's experience in a context
CIE coordinate	a standard-observer colour match
screen colour	a coded request to three device primaries
category	a word used by a community for a region of appearances

These descriptions can refer to related cases. They are not synonyms.

Goals

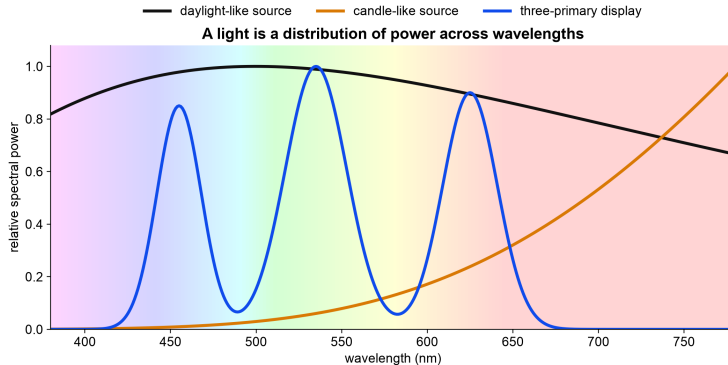
1. Distinguish wavelength, spectrum, cone response, appearance, coordinate, and code.
2. Explain univariance and metamerism without the “three colour buckets” overclaim.
3. Construct and read the CIE 1931 xy diagram.
4. Explain why a three-primary display has a triangular gamut.
5. Turn the model into a small Streamlit application.



PART I

Start with the light

A wavelength is not a complete description of a light



Course calculation. Black-body curves are illustrative; display peaks are schematic.

A light is a function

$$P(\lambda) = \text{power at wavelength } \lambda$$

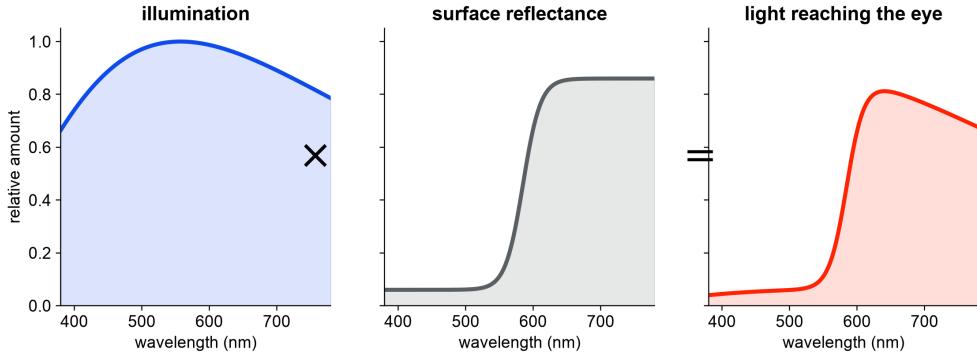
A spectral power distribution contains one value for every wavelength sampled.

- a narrow peak: approximately monochromatic;
- a broad curve: many wavelengths;
- three peaks: a typical display mixture;
- one colour word: none of the above is specified.

The spectrum is the physical input. Colour appearance is a result downstream.

An object does not send a fixed colour signal

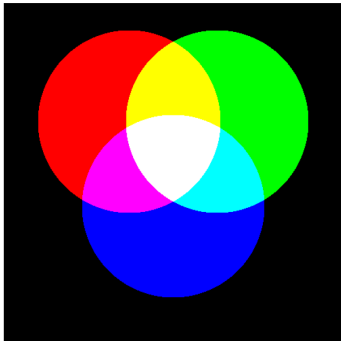
For an ordinary surface: illumination $\times$ reflectance = received spectrum



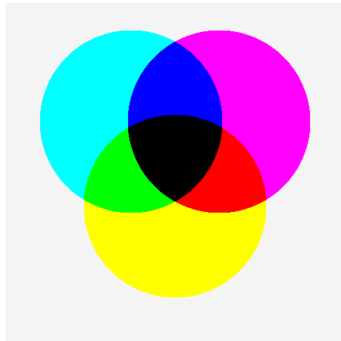
Relative teaching model: received spectrum = illuminant $\times$ surface reflectance.

“Mixing colour” names different operations

Add light: channel powers add



Stack ideal filters: transmissions multiply



Additive mixing combines emitted spectral power. Ideal subtractive mixing multiplies transmissions; real pigments also scatter and interact.

Four statements that should now sound different

- “This source has a peak at 470 nm.”
- “This surface reflects more short-wavelength light under this lamp.”
- “This patch appears blue in this surround.”
- “This pixel is encoded as #0068B7 in sRGB.”

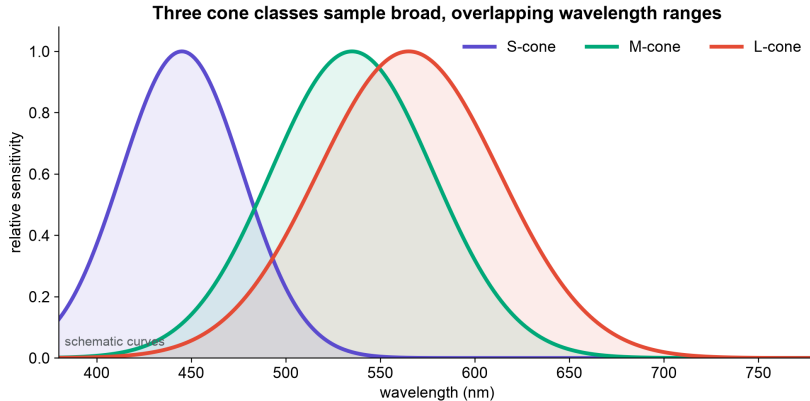
Only the first statement assigns a wavelength directly to the light.
None alone specifies the other three.



PART II

What the eye keeps — and loses

Three cone classes sample overlapping ranges



Schematic curves for teaching. S, M, and L mean short-, middle-, and long-wavelength sensitive.

One cone cannot identify a wavelength

$$q_i = \int P(\lambda) s_i(\lambda) d\lambda$$

Multiply the incoming spectrum by one sensitivity curve, then add over wavelength.

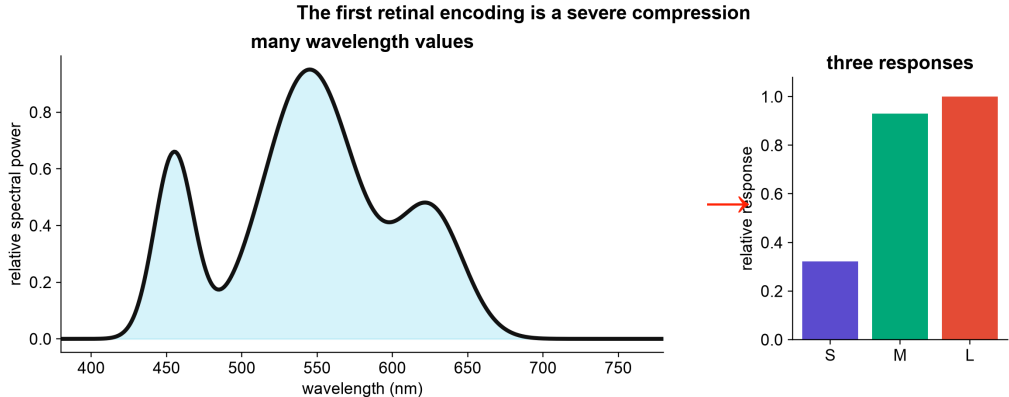
This is the principle of univariance.

The same response can be caused by:

- a weak light near peak sensitivity;
- a stronger light farther from the peak;
- many mixtures across the spectrum.

Once absorbed, the photon carries no wavelength label in that cone's output.

A spectrum is compressed to three responses



The bucket analogy, repaired

Useful part

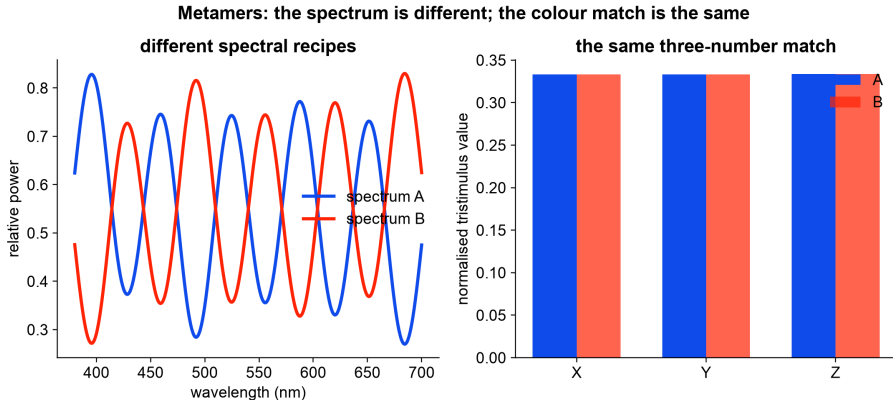
- three buckets have different filters;
- each produces only one total;
- the original spectrum cannot be reconstructed from three totals.

Where it breaks

- cone sensitivities overlap broadly;
- responses adapt and saturate;
- later neurons compare cones;
- context and spatial structure alter appearance.

Three cone classes explain the dimensionality of colour *matching*. They do not, by themselves, explain all colour appearance.

Different spectra can produce the same match



Course construction using the CIE 1931 colour-matching functions.

Metamerism is the display bargain

A screen does not reconstruct the original spectral recipe. It emits a new spectrum from three primaries that is intended to produce the required match.

What this makes possible

- colour photography;
- television and computer displays;
- compact three-channel specifications.

Where a match can fail

- a different observer;
- a different illuminant;
- fluorescence;
- changed adaptation or field size.

The same code can look different

The two centre patches contain identical RGB values



Identical centre RGB values do not guarantee identical appearance. The visual system compares each patch with its surround.

After the cones, signals are compared

luminance-like $L + M$
red–green-like $L - M$
blue–yellow-like $S - (L + M)$

These are schematic opponent combinations, not a complete neural model. They make one point clear: the brain does not receive an RGB image from the retina.

Adaptation, local contrast, spatial scale, and viewing sequence all enter before conscious appearance.



PART III

How colour became measurable

Different thinkers solved different problems

Work	Object	Question
Newton, 1704 Goethe, 1810	rays and prisms appearance	How does white light separate and recombine? How do boundaries, shadows, afterimages, and context affect experience?
Maxwell, 1850s–60s Munsell, 1905	matches surface samples	How can three primaries reproduce a match? How can hue, value, and chroma be ordered for practice?
CIE, 1931	standard observer	How can colour matches be reported reproducibly?

A later framework does not turn every earlier question into the same question.

Newton and Goethe attend to different layers

Newton

- controlled the geometry;
- separated light by refraction;
- showed white light can be compound;
- described the stimulus.

Goethe

- studied coloured shadows;
- recorded afterimages and boundaries;
- foregrounded the observer;
- described appearance in context.

Goethe's physical theory did not replace Newton's optics. His observations still warn us not to confuse a spectrum with an appearance.

Colour matching came before direct cone measurement

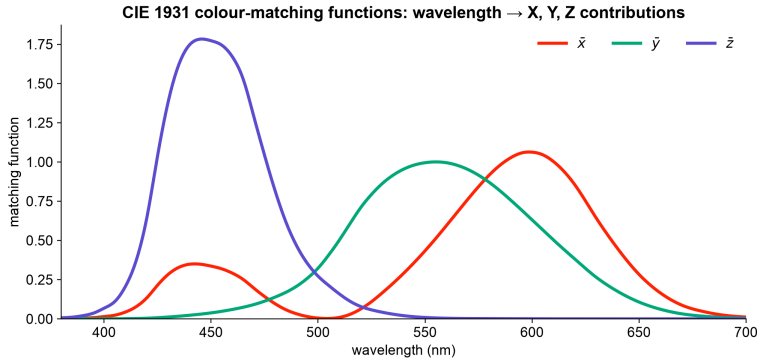
An observer adjusted three primary lights until one half of a field matched a test light in the other half.

Some wavelengths could not be matched using positive amounts of the chosen real primaries. One primary was moved to the test side:

$$\text{test} + rR = gG + bB \quad \Longleftrightarrow \quad \text{test} = (-r)R + gG + bB.$$

The negative value records the experimental operation. It is not “negative light.”

CIE XYZ is a mathematical coordinate system



XYZ is a linear transform of colour-matching results. X, Y, and Z are not physical lamps and not the three cone signals; Y was chosen to carry standard photopic luminance.

From a spectrum to three CIE values

$$X = \int P(\lambda) \bar{x}(\lambda) d\lambda$$

$$Y = \int P(\lambda) \bar{y}(\lambda) d\lambda$$

$$Z = \int P(\lambda) \bar{z}(\lambda) d\lambda$$

Change the spectrum and recalculate. Any spectra with the same XYZ values are a standard-observer colour match under the specified conditions.

Chromaticity removes one scale factor

$$x = \frac{X}{X + Y + Z}, \quad y = \frac{Y}{X + Y + Z}, \quad z = 1 - x - y$$

- Multiplying X, Y, and Z by the same amount leaves x and y unchanged.
- Two fractions are enough because the three proportions sum to one.
- xy omits overall tristimulus scale. It is not a complete colour appearance.

Build the spectral locus one wavelength at a time

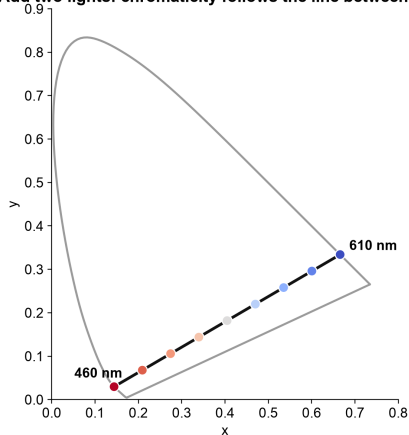
For each row in the official CIE table:

1. take the wavelength's $\bar{x}$, $\bar{y}$, and $\bar{z}$ values;
2. divide by their sum;
3. plot (x, y) ;
4. repeat for the next wavelength.

The curved path is the **spectral locus**: the route taken by monochromatic lights through the xy diagram.

Adding two lights traces a straight line

Add two lights: chromaticity follows the line between them

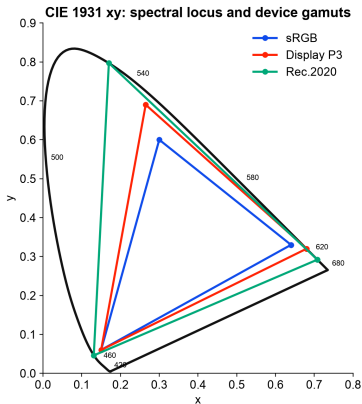


Why real-light mixtures lie inside

- spectra add;
- XYZ values therefore add;
- after normalisation, a two-light mixture lies between the two chromaticities;
- a spectrum with many wavelengths is a non-negative mixture of many points.

The straight closure between the red and violet ends is the **line of purples**. Purple lies there because it is produced by mixing the ends; it is not a single wavelength.

Three physical primaries make a triangle



CIE 1931 2° data; primary coordinates from sRGB/Rec.709, Display P3, and BT.2020 specifications.

The horseshoe is not a screenshot of every visible colour

- The outline represents standard-observer chromaticity coordinates.
- The page is itself shown through your display's primaries and calibration.
- Points outside the display gamut can be plotted as locations but not reproduced faithfully.
- Different luminances share the same xy point.
- Equal distances in xy are not equal perceived differences.



PART IV

Use colour as an encoding — not decoration

Hue, lightness, and chroma do different jobs

Dimension	Useful for	Do not assume
hue	category, identity, highlighting	a natural numeric order
lightness	ordered magnitude if mon- tonic	equal code steps look equal
chroma	emphasis, status, coarse order	precise quantitative comparison

Let position or length carry values that must be compared precisely. Use colour to organise, order coarsely, or direct attention.

A palette is viewed as a scene

- test colours on the actual background;
- test them at the actual mark and text sizes;
- inspect common colour-vision deficiencies;
- make lightness monotonic in sequential scales;
- reserve diverging palettes for a meaningful midpoint;
- add labels, position, shape, or line style when the distinction matters.

The viewer experiences relationships between marks, not hexadecimal values in isolation.



PART V

Make the model interactive with Streamlit

Streamlit has a small mental model



Whenever a widget changes, Streamlit reruns the script from top to bottom with the new value.

One widget turns a constant into an interface

Notebook or script

```
wavelength = 550  
print(wavelength)
```

Streamlit

```
import streamlit as st  
wavelength = st.slider(  
    "Wavelength (nm)", 380, 700, 550  
)  
st.write(wavelength)
```

```
python -m streamlit run hello_colour.py
```

The lab builds three linked views

1. **Spectrum:** change peak wavelength and spectral width.
2. **Eye:** integrate the spectrum against three schematic cone curves.
3. **Chromaticity:** use the official CIE table to locate the light in xy.

Final extension: add an intensity slider. Cone-response magnitudes should change; chromaticity should remain almost fixed.

Keep the boundaries visible

- Cone curves in the app are schematic.
- CIE matching functions are standard-observer data, not cone sensitivities.
- A wavelength rendered on screen is an RGB approximation, not monochromatic light.
- xy omits luminance and viewing context.
- The app is an explanatory model, not a calibrated colour-management system.

Exit ticket

Complete the chain for one colour in a chart:

The display emits _____. The eye reduces this to _____.
The viewer perceives it in relation to _____.
The design uses that appearance to encode _____.

Name one thing lost at each translation.

Sources and data

- CIE (2019), *CIE 1931 colour-matching functions, 2 degree observer*, DOI: 10.25039/CIE.DS.xvudnb9b.
- Kalloniatis, M. and Luu, C., "The Perception of Color," *Webvision*, NCBI Bookshelf.
- Newton, I. (1704), *Opticks*; Goethe, J. W. von (1810), *Theory of Colours*.
- Munsell, A. H. (1905), *A Color Notation*.
- W3C, *CSS Color Module Level 4*; IEC 61966-2-1 (sRGB); ITU-R BT.709 and BT.2020.
- Streamlit, "Basic concepts" and API documentation.

All plotted figures are course calculations. Official CIE data are supplied with their metadata under CC BY-SA 4.0.